

Capacitor Charging and a Power On Reset for an Active Low Microcontroller

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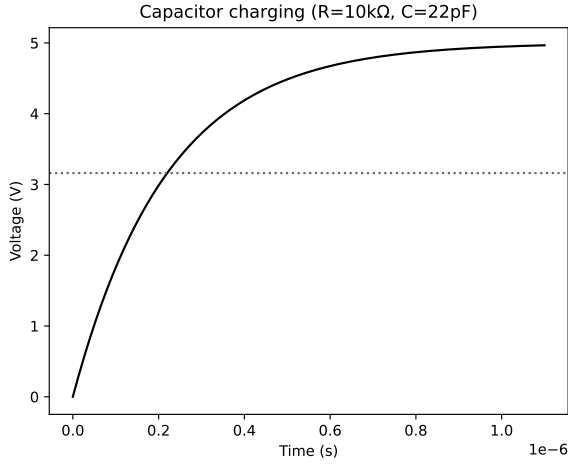


Fig. 1. Charging curves for four resistances, $C = 22$ pF.

Abstract—This document derives the charging behaviour of a resistor-capacitor (RC) network, the voltage reached after one time constant, and applies the result to size a power-on-reset (POR) circuit for a microcontroller with an active-low reset.

I. INTRODUCTION

When a capacitor C is charged through a series resistor R from a constant supply V_0 , the capacitor voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right), \quad (1)$$

where $\tau = RC$ is the time constant. Substituting $t = \tau$ gives $V(\tau) = V_0(1 - e^{-1}) \approx 0.632 V_0$.

II. POWER ON RESET WITH AN ACTIVE LOW RESET

A power-on-reset circuit exploits (1) directly: R from V_{CC} to the reset node and C from that node to ground hold the microcontroller's active-low reset pin low until the node crosses V_{IH} , releasing the device.

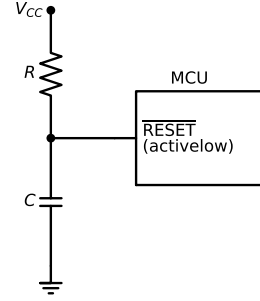


Fig. 2. Power-on-reset network driving an active-low reset pin.

A. Sizing R for a Timing Constraint

The reset pin releases the instant the node voltage reaches V_{IH} . Setting $V(t_{\text{release}}) = V_{IH}$ in (1) and solving for t_{release} :

$$\begin{aligned} V_{IH} &= V_0 \left(1 - e^{-t_{\text{release}}/RC}\right) \\ \frac{V_{IH}}{V_0} &= 1 - e^{-t_{\text{release}}/RC} \\ e^{-t_{\text{release}}/RC} &= 1 - \frac{V_{IH}}{V_0} \\ -\frac{t_{\text{release}}}{RC} &= \ln \left(1 - \frac{V_{IH}}{V_0}\right) \\ t_{\text{release}} &= -RC \ln \left(1 - \frac{V_{IH}}{V_0}\right). \end{aligned} \quad (2)$$

Rearranging (2) for R , given a target t_{release} and a chosen C :

$$R = \frac{-t_{\text{release}}}{C \ln \left(1 - \frac{V_{IH}}{V_0}\right)}. \quad (3)$$

1) *Worked example:* Take $V_{IH} = 0.7 V_0$, $V_0 = 5.0$ V, and a target $t_{\text{release}} = 10$ ms. The $C = 22$ pF used for the plots above was chosen only to keep the curves readable at a sub-microsecond timescale; a practical timing design instead starts from $C = 100$ nF. Evaluating the denominator of (3):

$$\ln(1 - 0.7) = \ln(0.3) \approx -1.204. \quad (4)$$

Substituting into (3):

$$R = \frac{-(10 \times 10^{-3})}{(100 \times 10^{-9}) \times (-1.204)} \approx 83 \text{ k}\Omega. \quad (5)$$

An 83 k Ω resistor with a 100 nF capacitor gives a reset hold-off of approximately 10 ms. Keeping C at the plotting value of